

Ahmed M. Elkholy

Electrical Power Engineer | Assistant Lecturer | PhD Student

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Profile

Electrical power engineer and Assistant Lecturer at Tanta University, pursuing a PhD at Moscow Power Engineering Institute (MPEI), with expected completion in December 2026. Research focuses on distribution network modelling, power quality, FACTS devices, fault mitigation, and electric vehicle charging infrastructure. Experience includes university teaching, power system simulation, and optimization.

Academic and Research Experience

Researcher / R&D Engineer — Moscow Power Engineering Institute, Russia | October 2023–present

- Conduct research in the Industrial Electronics Department on distribution systems, power electronic devices, and network performance.
- Develop and evaluate models and optimization methods for power quality improvement, fault mitigation, and EV charging infrastructure.

Assistant Lecturer — Faculty of Engineering, Tanta University, Egypt | July 2019–present

- Prepare teaching materials and deliver tutorials and laboratory sessions in electrical engineering.
- Support assessment preparation and marking; guide students in course and graduation projects.
- Conduct research in power system analysis, control, and renewable energy integration.

Demonstrator — Faculty of Engineering, Tanta University, Egypt | March 2016–July 2019

- Assist students with laboratory exercises, practical work, and equipment preparation.
- Provide feedback on student performance and support assessment marking and laboratory safety.

Education

PhD studies in Electrical Engineering — Moscow Power Engineering Institute, Russia | In progress; expected December 2026

MSc in Electrical Power and Machines Engineering — Tanta University, Egypt | 2016–2019

Thesis: *Islanding Scenarios for High Reliable Operation of Distribution Networks*. Supervisors: Prof. Ahmed Refat Azmy and Dr. Hossam Salah.

BSc in Electrical Power and Machines Engineering — Tanta University, Egypt | 2009–2014

Graduation project: *The Impact of Large Wind Energy Penetration on the Performance of Electric Power Systems*. Supervisor: Prof. Ahmed Refat Azmy. Project grade: Excellent (99%).

Technical Skills and Research Software

- **Power systems:** distribution network modelling, unbalanced power flow, symmetrical components, power quality, FACTS devices, islanding, load shedding, and renewable energy integration.
- **Modelling and programming:** MATLAB/Simulink, Python, SQL, Git/GitHub, and LaTeX; genetic algorithms and numerical optimization.
- **Research software:** [IEEE13-optimization](#), for unified power controller optimization in the IEEE 13-bus network; [Zthevenin](#), a MATLAB tool for calculating power system Thevenin impedance.

Training, Service, and Languages

- Programming for Data Science for Enterprise Nanodegree, Udacity, 2019: Python, SQL, and Git.
- Faculty and Leadership Development Centre, Tanta University, 2019: teaching technology, presentation skills, research project preparation, crisis management, and teaching quality.
- PLC training, JELECOM, 2013; IoT Foundations: Standards and Ecosystems, LinkedIn Learning, 2020.
- Reviewer and conference organization support, MEPCON 2019.
- Egyptian Engineers Syndicate member since 2014.
- **Languages:** Arabic (native), English (professional working proficiency), Russian (basic to intermediate).

1. A. M. Elkholy, D. I. Panfilov, A. N. Rozhkov, A. V. Badalyan, I. A. Cherdintsev. **Statistically robust genetic algorithm planning of EV fast-charging corridors under grid and storage constraints.** *Electric Power Systems Research*, 2026, vol. 261, article 113505 (online publication; December issue). DOI
2. D. I. Panfilov, M. G. Astashev, A. V. Chasov, A. M. Elkholy. **Implementation of Shunt Regulation Devices for Balancing the Operating Modes of Four-Wire Networks with Asymmetric Loads.** *Russian Electrical Engineering*, 2026, vol. 97, no. 7, pp. 589–595. DOI
3. M. G. Astashev, D. I. Panfilov, S. N. Demidov, V. V. Mikryukov, A. V. Chasov, A. M. Elkholy. **Modeling and Analysis of Distribution Networks with Asymmetric Loads under Dual-Loop Shunt Regulation of Their Operating Modes.** *Russian Electrical Engineering*, 2026, vol. 97, no. 7, pp. 581–588. DOI
4. A. M. Elkholy, D. I. Panfilov, M. G. Astachev. **Arc Suppression Devices for Mitigating Single Line-to-Ground Faults and Enhancing Power System Resilience.** *IEEE Transactions on Industry Applications*, 2026, vol. 62, no. 1, pp. 596–607. DOI
5. A. M. Elkholy, A. N. Rozhkov, A. V. Badalyan, I. A. Cherdintsev. **Adaptive genetic algorithms enhance EV charging infrastructure resilience through multi-constraint optimization of grid resources and traffic dynamics.** *Electric Power Systems Research*, 2026, vol. 250, article 112045. DOI
6. A. M. Elkholy, A. V. Chasov, D. I. Panfilov. **Statistical approach to non-invasive modeling of zigzag transformers using variable impedance.** *e-Prime - Advances in Electrical Engineering, Electronics and Energy*, 2025, vol. 14, article 101135. DOI
7. R. N. Krasnoperov, D. I. Panfilov, M. G. Astachev, A. M. Elkholy. **Performance evaluation of a low-voltage SVC utilizing IoT streamed data for distribution systems.** *Electric Power Systems Research*, 2025, vol. 239, article 111187. DOI
8. A. M. Elkholy, D. I. Panfilov, A. E. ELGebaly. **Comparative analysis of FACT devices for optimal improvement of power quality in unbalanced distribution systems.** *Scientific Reports*, 2025, vol. 15, article 2672. DOI
9. A. M. Elkholy, D. I. Panfilov, M. G. Astashev, A. V. Chasov, I. D. Panfilov, V. N. Ostrirov. **Analysis of the Influence of Symmetrical Devices on the Operating Modes of Distribution Networks.** *Russian Electrical Engineering*, 2024, vol. 95, no. 6, pp. 425–440. DOI
10. Ahmed M. Elkholy, D. I. Panfilov, M. G. Astashev. **The Influence of Transmission-Line Parameters and Its Load on Losses and Power Quality at the Load-Connection Point.** *Russian Electrical Engineering*, 2023, vol. 94, no. 6, pp. 403–411. DOI
11. A. M. Elkholy, I. B. M. Taha, S. Kamel, M. K. El-Nemr. **Grid Synchronization Enhancement of Distributed Generators Using an Adaptive Phase-Locked Loop Tuning System.** *Electronics*, 2022, vol. 11, no. 5, pp. 702. DOI
12. A. M. Elkholy, H. A. Abd el-Ghany, A. M. Azmy. **General Framework for Intentional Islanding to Enhance Distribution System Performance.** *Electric Power Components and Systems*, 2020, vol. 48, no. 14-15, pp. 1488–1507. DOI

Conference Publications

REEPE: Radio Electronics, Electrical and Power Engineering. EDM: Electron Devices and Materials. MEPCON: Middle East Power Systems Conference.

13. I. A. Cherdintsev, A. N. Rozhkov, A. V. Badalyan, A. M. Elkholy. **Influence of Frequency and Phase on the Operation and Resonance of the DC-DC Converter.** *IEEE REEPE*, 2026, pp. 1–6. [DOI](#)
14. A. M. Elkholy, A. N. Rozhkov, A. V. Badalyan, I. A. Cherdintsev. **Genetic Algorithm-Based Optimization of EV Charging Station Placement on Long-Distance Routes.** *IEEE REEPE*, 2025, pp. 1–7. [DOI](#)
15. A. M. Elkholy, D. I. Panfilov, M. G. Astashev, A. V. Chasov. **Comprehensive Study of the Impact of Unbalanced Loading on Power Losses and Voltage Quality in Distribution Systems.** *IEEE EDM*, 2024, pp. 1340–1346. [DOI](#)
16. A. M. Elkholy, D. I. Panfilov, M. G. Astachev. **Enhancing Power System Resilience: An Analysis of Novel Arc Suppression Device Technology in Mitigating Single Line to Ground Faults.** *IEEE REEPE*, 2024, pp. 1–7. [DOI](#)
17. A. M. Elkholy, D. I. Panfilov, M. G. Astachev. **Assessment of Various Technologies Influence to Reduce the Impact of Single-Phase Earth Faults in Distribution Systems with an Isolated Neutral.** *IEEE REEPE*, 2024, pp. 1–8. [DOI](#)
18. A. M. Elkholy, D. I. Panfilov, M. G. Astashev. **Design and Implementation of an Arc Suppression Device for Single Line to Ground Faults in Power Systems.** *IEEE MEPCON*, 2023, pp. 1–7. [DOI](#)
19. A. M. Elkholy, D. I. Panfilov, A. E. El Gebaly. **Evaluation of Zero and Negative Sequence Currents Influence of Asymmetric Load on the Power Losses and Quality in Distribution Networks.** *IEEE EDM*, 2023, pp. 1110–1115. [DOI](#)
20. A. M. Elkholy, D. I. Panfilov, A. E. ELGebaly. **Identifying the Rating of the Unified Power Controller According to Unbalanced Distribution System Requirements.** *IEEE EDM*, 2023, pp. 960–965. [DOI](#)
21. D. I. Panfilov, A. M. Elkholy, A. E. ELGebaly. **Evaluation of the Influence of AC Voltage Regulators, SVC and Unified Power Controllers on Power Losses and Quality in Unsymmetrical Distribution Systems.** *IEEE EDM*, 2023, pp. 950–956. [DOI](#)
22. M. G. Hussien, A. M. Elkholy, D. A. Mansour, M. M. Akl, M. Numair, I. A. Soliman. **Performance Enhancement of Wave Energy Converters Integrated with DC Grids Using Energy Storage Systems.** *IEEE MEPCON*, 2021, pp. 77–83. [DOI](#)
23. I. A. Soliman, M. Numair, M. M. Akl, D. A. Mansour, A. M. Elkholy, M. G. Hussien. **Hosting Capacity Enhancement Through IoT-based Active Power Curtailment of PV Generation.** *IEEE MEPCON*, 2021, pp. 259–264. [DOI](#)
24. A. M. Elkholy, H. A. A. El-Ghany, A. M. Azmy. **An Advanced Load Shedding Algorithm to Enhance Intentional-Islanding Dynamics.** *IEEE MEPCON*, 2018, pp. 797–802. [DOI](#)
25. A. M. Elkholy, H. A. A. El-Ghany, A. M. Azmy. **A proposed load shedding mechanism for enhancing intentional-islanding dynamics of distribution systems.** *IEEE MEPCON*, 2017, pp. 870–875. [DOI](#)